THE PANDEMIC- TRENDS IN COVID-19 VARIANTS

dinner pool

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AGENDA

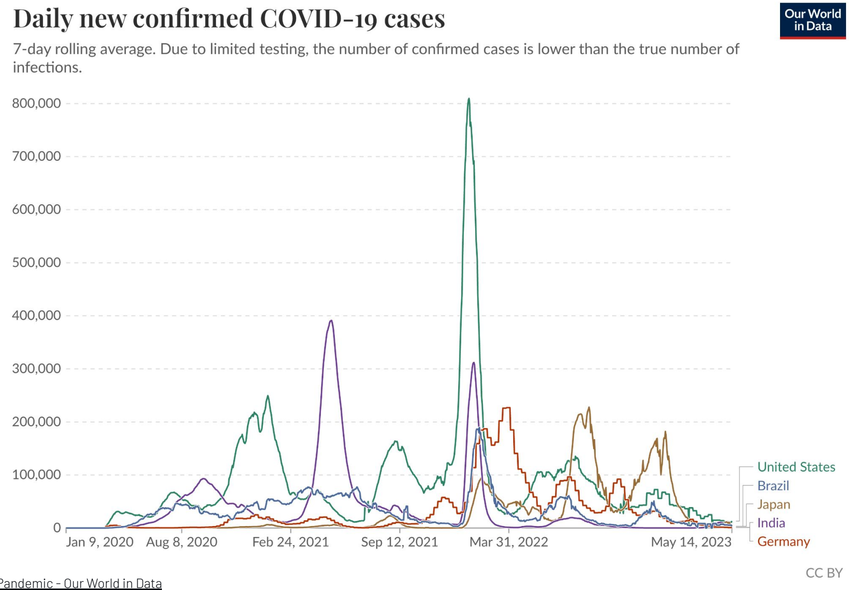
Introduction to COVID-19

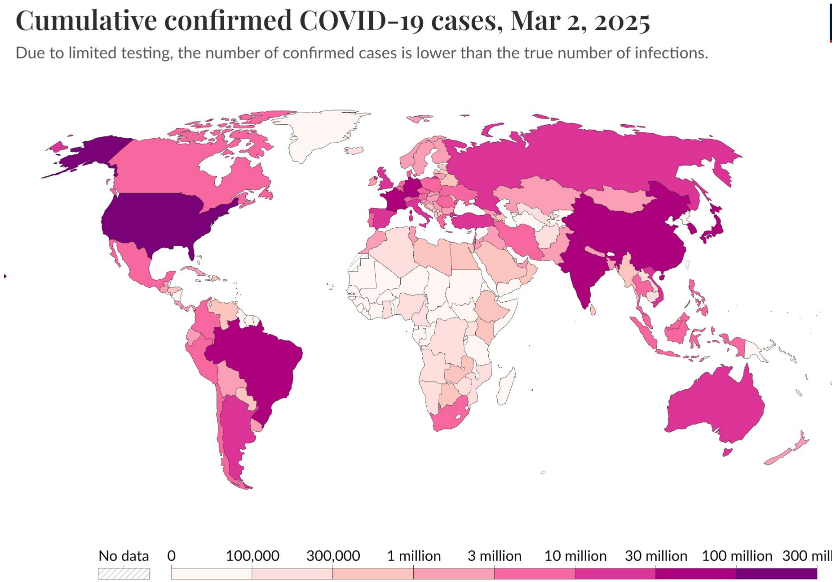
- COVID-19 is an infectious disease caused by the **SARS-CoV-2 virus.**
- It spreads primarily through **respiratory droplets** from coughing, sneezing or talking
- Transmission can also occur through contact with contaminated surfaces.
- The disease can cause respiratory illness, long-term health complications, and severe cases may lead to hospitalization or death.

- The first cases appeared in **Wuhan, China,** in December 2019.
- By January 2020, COVID-19 had spread worldwide, leading to a **global pandemic.**
- The most widely accepted theory suggests it originated from **animal-to-human** transmission, likely from bats through an intermediate host.

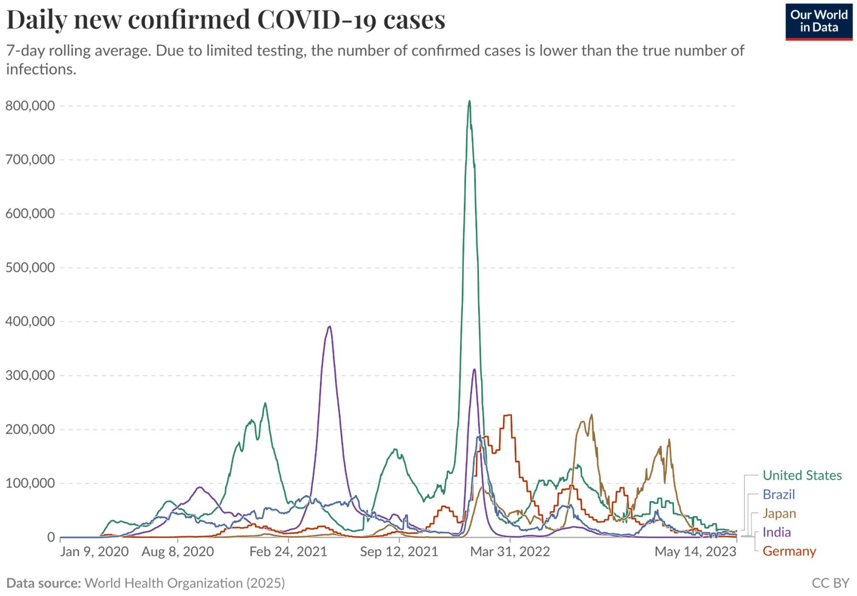
Analysis of COVID-19

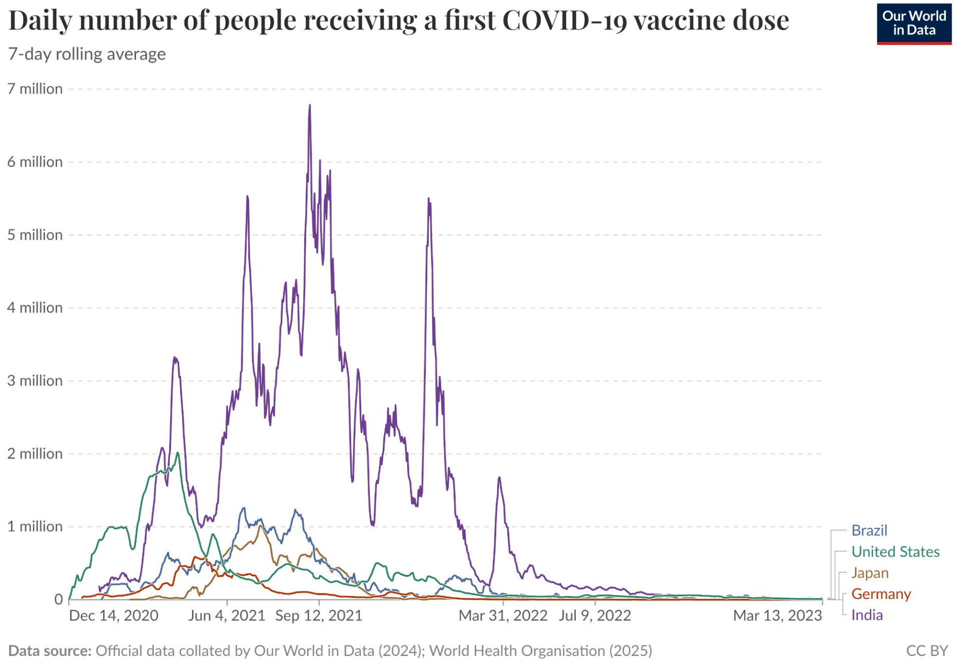
COVID-19 had a global reach, with countries like USA, Italy, India and China being severely affected. Its progress was in waves, with second wave being devastating in terms of cases and deaths.





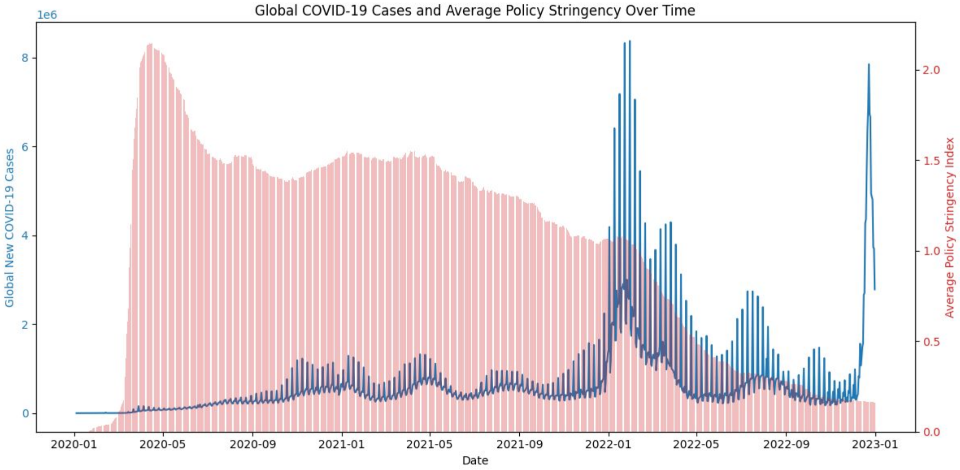
Progress of COVID-19 and Vaccination rates

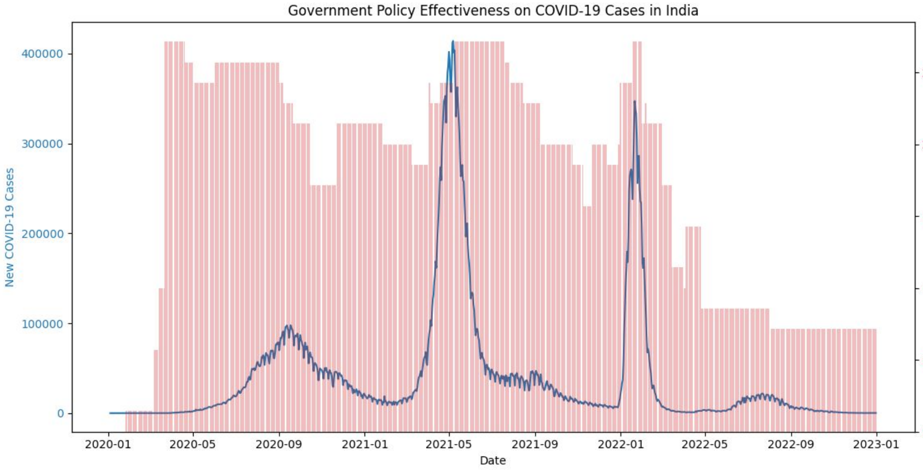




COVID-19 Pandemic - Our World in Data

COVID-19 and Government Policies





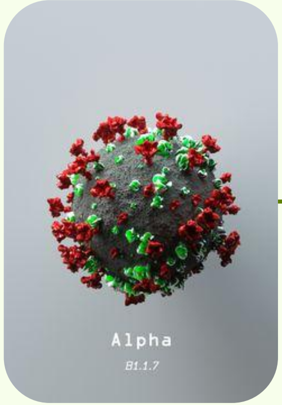
Why Study Variants?

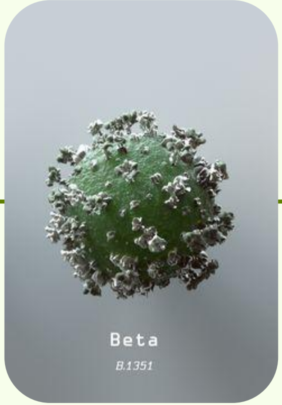
- SARS-CoV-2 is prone to mutations and the generation of genetic variants.
- This leads to new variants with possibly higher levels of transmission, severity, and immune evasion.
- Subsequent variants can spread faster, cause more severe illness, reduce vaccine effectiveness, or produce false negative results on the diagnostic tests.
- Genomic surveillance helps track variants to inform public health responses and control outbreaks.

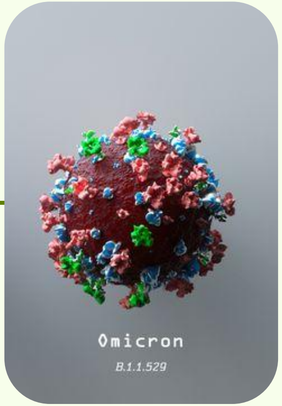
Variant Classification

According to WHO, variants of SARS-CoV-2 are classified into three main categories based on their potential impact on public health:

1. **Variant under Monitoring (VUM)** - Requires close observation to assess its potential threat to global health.
2. **Variant of Interest (VOI)** - Variant with changes that have increased risk to global population.
3. **Variant of Concern (VOC)** - A VOI that also has increased severity, decreased options of treatment, and can also hinder the performance of health systems. VOC's are given a Greek symbol , for example: Alpha, Beta, Delta







Alpha (B.1.1.7)

Beta (B.1.351)

Gamma (B.1.1.28.1)

Delta (B.1.617.2)

Omicron (B.1.1.529)

Key Variants

Alpha, Delta, Gamma, Beta, and Omicron were designated as key variants of concern (VOCs) due to their increased ***transmissibility***, and/or increased ***severity***, leading to significant global impact on the COVID-19 pandemic

Understanding COVID-19 Variants

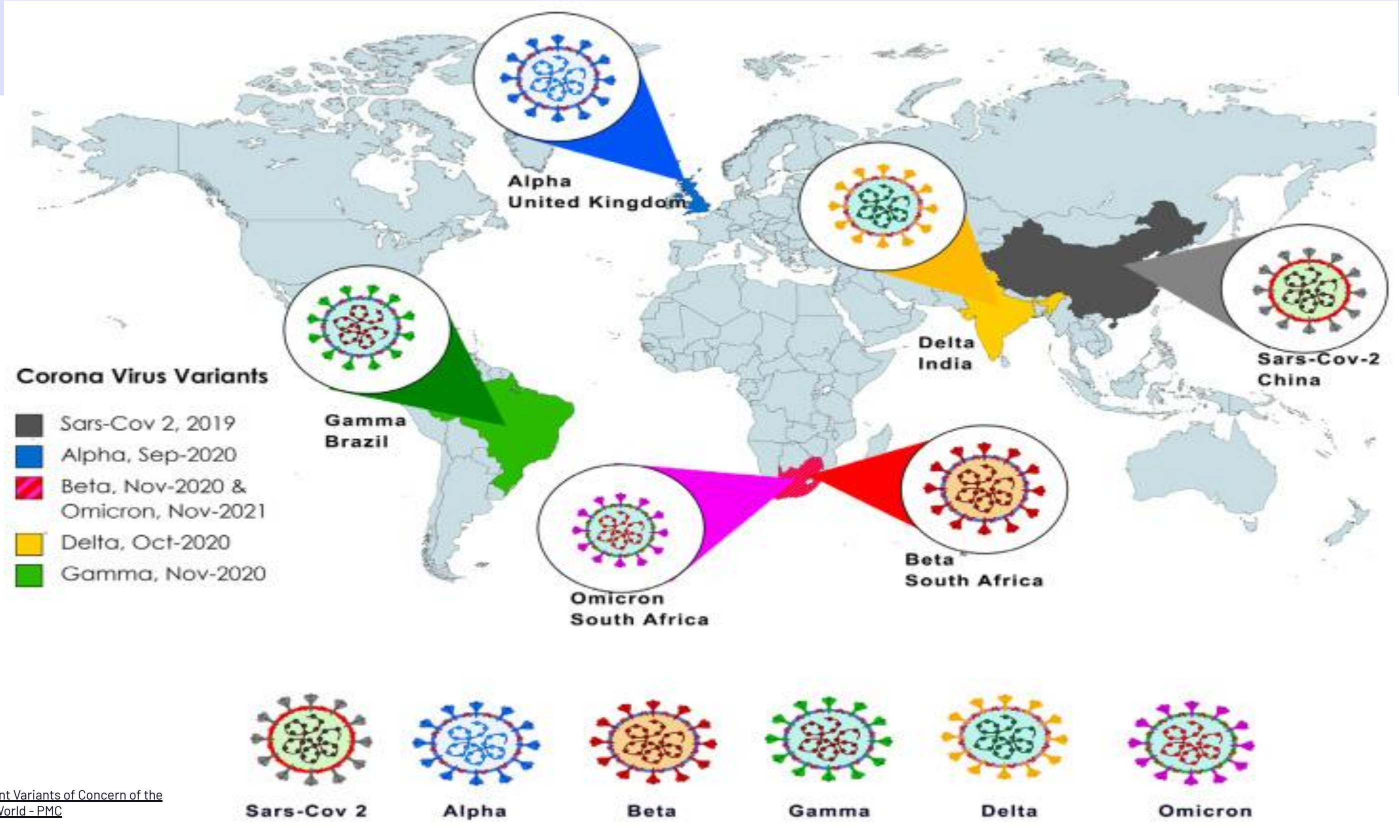
Mutation refers to a change in the viral genetic material (DNA or RNA) that can lead to new traits.

Variants arise from different lineages with some shared mutations due to selection pressure.

Variants with advantageous mutations replace earlier strains over time.

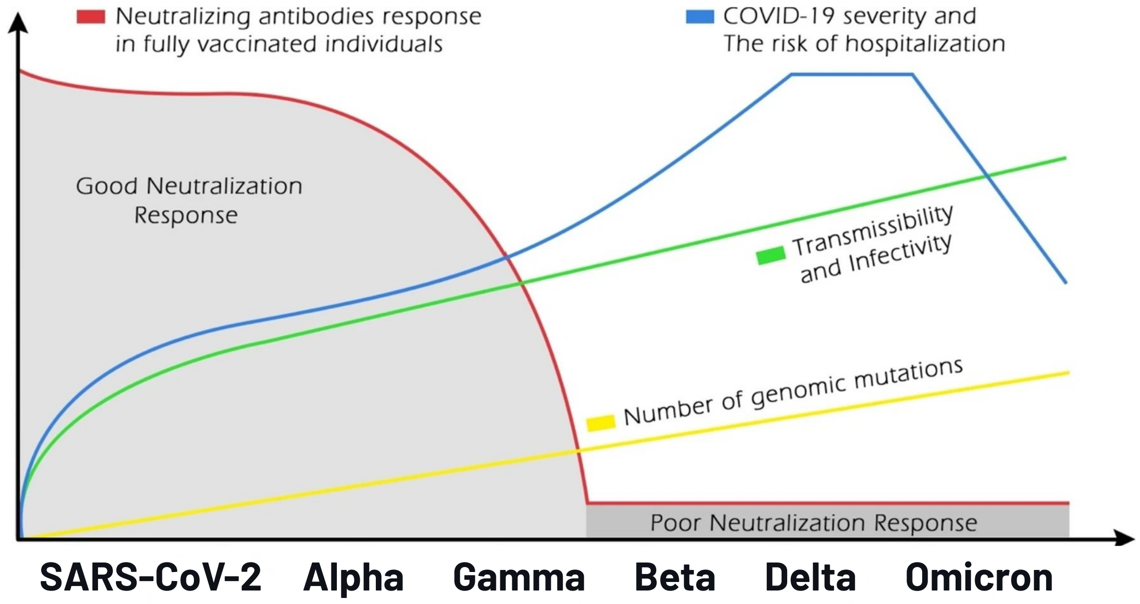
Alpha →	Beta & Gamma →	Delta →	Omicron
First major variant, more transmissible than original strain.	Improved immune evasion, reducing vaccine effectiveness.	Highly transmissible, causing severe disease and global surges.	Extensive mutations led to high immune escape but lower severity.

Timeline of Key Variants



From Alpha to Omicron: How Different Variants of Concern of the SARS-Coronavirus-2 Impacted the World - PMC

Analysis of Key Variants



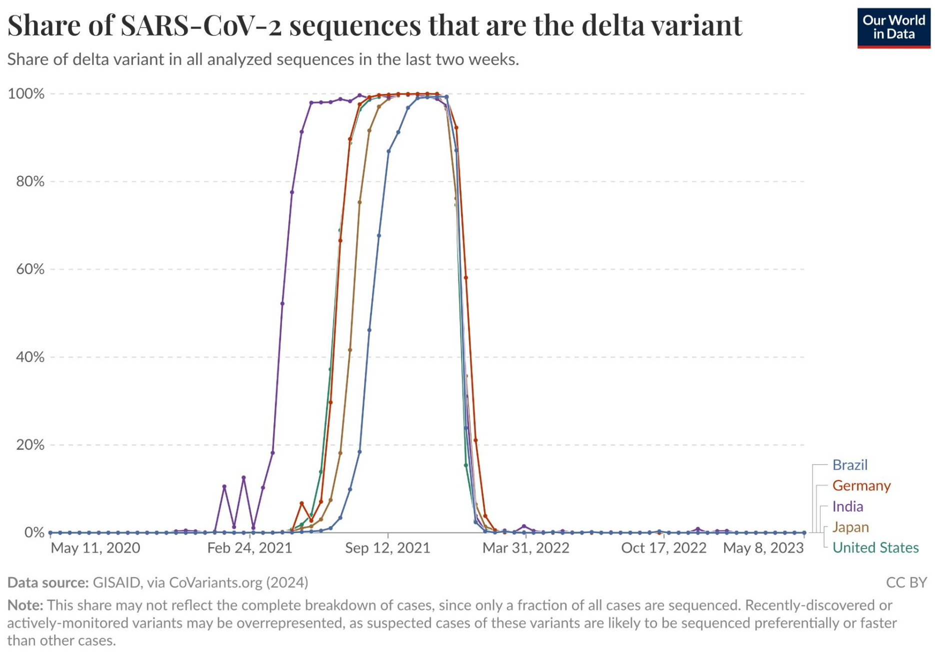
Rising mutations increased transmissibility, with Delta peaking in hospitalizations before Omicron's lower severity.

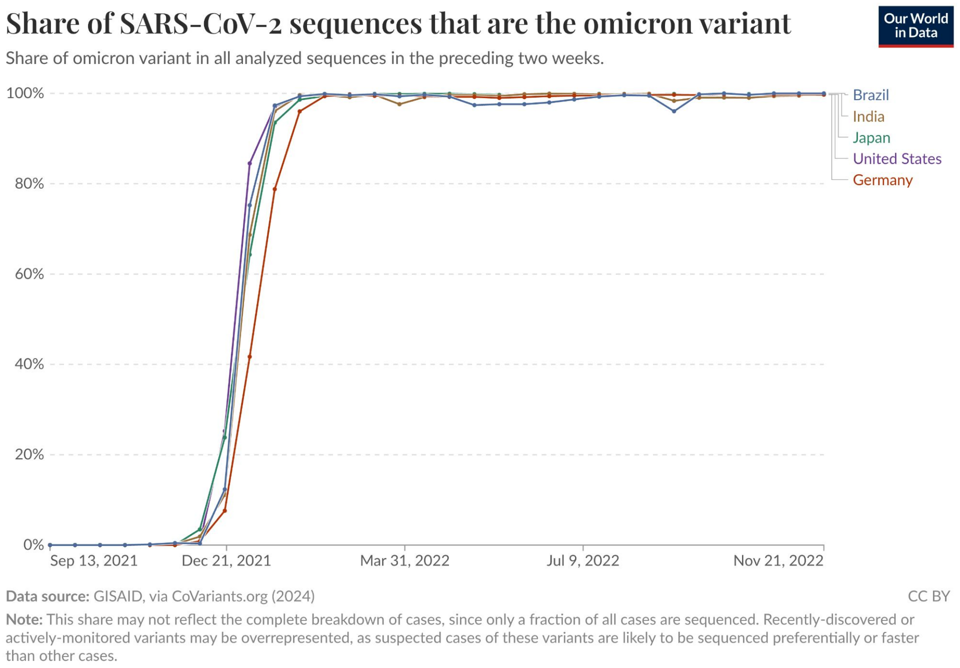
Vaccine efficacy declined as variants like Beta and Delta evaded immunity.

Boosters and updated vaccines helped counteract evolving variants.

SARS-CoV-2 journey: from alpha variant to omicron and its sub-variants | Infection

Rise of Delta & Omicron



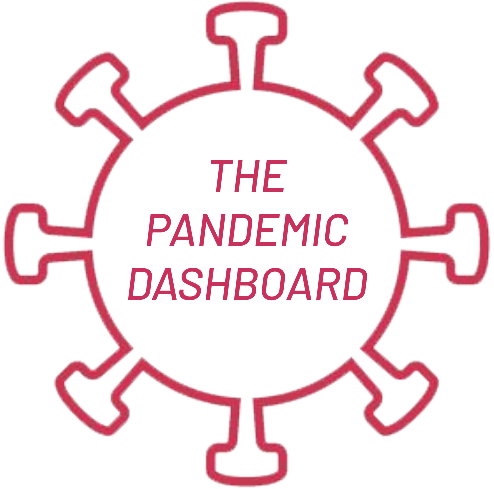


COVID-19 Pandemic - Our World in Data

Our Dashboard

COVID-19 Variant Trends

SEI^{3}RD Simulation



Seasonality & Patterns

Forecasting the Next Wave

About Trends of COVID-19 Variants

Time Series Evaluation

Tracks how different variants rose and declined over time using case counts.

Helps in identifying peaks and transitions and how long a variant remains dominant.

Seasonal Emergence of Variants

Analyzes seasonal patterns in variant emergence, focusing on global trends vs. India's specific patterns.

Informs region-specific preparedness strategies.

STL Decomposition

Breaks down time series data into:

- Trend (long-term growth)
- Seasonality (recurring patterns)
- Residual (random fluctuations)

Distinguishes real growth from short-term fluctuations.

Forecasting using ARIMA

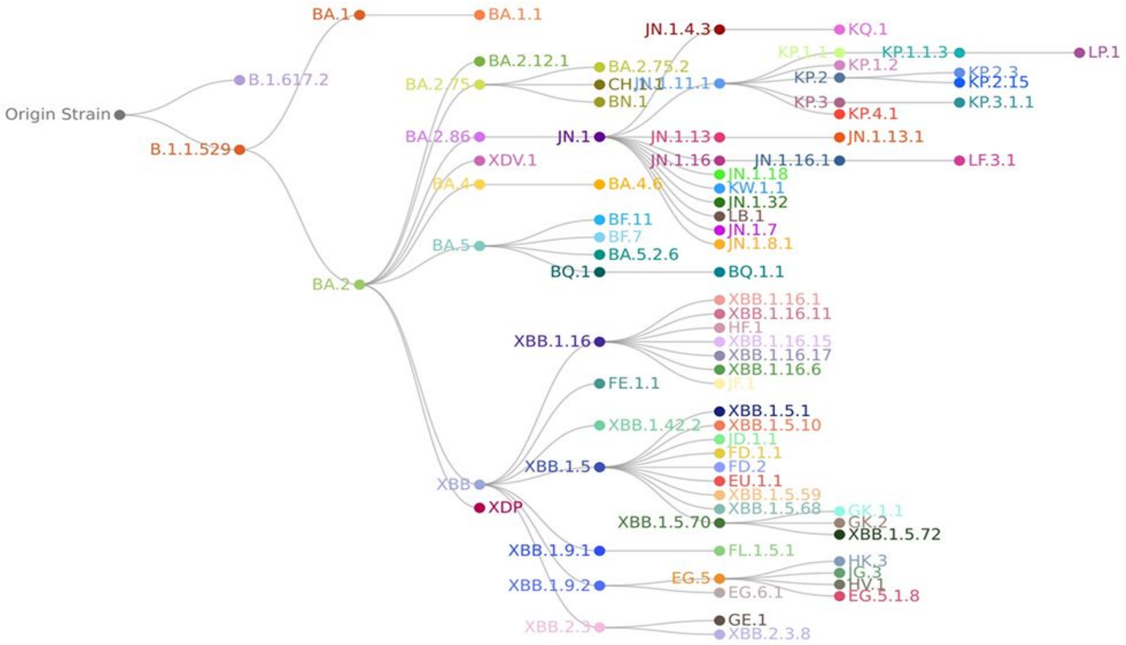
Predicts future variant trends based on past data using AutoRegressive Integrated Moving Average.

Helps in forecasting upcoming waves and planning healthcare resources in advance.

About Mutations

Multiple mutations accumulate over time, creating diverse viral lineages, some of which become dominant variants with higher transmissibility.

By linking specific mutations to changes in immune response, researchers can predict vaccine effectiveness and guide updates to counter emerging variants.



Moderna and Pfizer win approval for KP.2 vaccine as KP.3 variants account for about half of current cases

About SEIR Models

These models use Ordinary Differential Equations (ODE) to simulate the progress of an infectious disease.

The simulation is dictated by the parameters used like ε, γ and β, which are useful for simulating a basic set-up for a disease.

Symbol	Explanation
S	Susceptible individuals
E	Exposed individuals in the latent period
I	Infectious individuals
R	Recovered individuals with immunity
β	Infection rate
$1/\gamma$	Average infectious period
N	Total population size
R_0	Basic reproduction number
$1/\varepsilon$	Average latent period

$$\frac{\mathrm{dS}}{\mathrm{dt}} = -\beta IS,$$

$$\frac{\mathrm{dE}}{\mathrm{dt}} = \beta IS - \varepsilon E,$$

$$\frac{\mathrm{dI}}{\mathrm{dt}} = \varepsilon E - \gamma I,$$

$$\frac{\mathrm{dR}}{\mathrm{dt}} = \gamma I,$$

Extensions of the SEIR model

About SEI^{3}RD Models

$$\frac{dS_k}{dt} = -\sum_{\ell=1}^{K} \left(\beta_{\ell k}^{asym} I_\ell^{asym} + \beta_{\ell k}^{sym} I_\ell^{sym} + \beta_{\ell k}^{sev} I_\ell^{sev}\right) S_k,$$

$$\frac{dE_k}{dt} = \sum_{\ell=1}^{K} \left(\beta_{\ell k}^{asym} I_\ell^{asym} + \beta_{\ell k}^{sym} I_\ell^{sym} + \beta_{\ell k}^{sev} I_\ell^{sev}\right) S_k - \varepsilon_k E_k,$$

$$\frac{dI_k^{asym}}{dt} = \eta_k \varepsilon_k E_k - \gamma^{asym} I_k^{asym},$$

$$\frac{dI_k^{sym}}{dt} = (1-\eta_k)(1-\nu_k)\varepsilon_k E_k - \gamma_k^{sym} I_k^{sym},$$

$$\frac{dI_k^{sev}}{dt} = (1-\eta_k)\nu_k \varepsilon_k E_k - \left((1-\sigma_k(t))\gamma_k^{sev\text{-}r} + \sigma_k(t)\gamma_k^{sev\text{-}d}\right) I_k^{sev},$$

$$\frac{dR_k}{dt} = \gamma^{asym} I_k^{asym} + \gamma_k^{sym} I_k^{sym} + (1-\sigma_k(t))\gamma_k^{sev\text{-}r} I_k^{sev},$$

$$\frac{dD_k}{dt} = \sigma_k(t)\gamma_k^{sev\text{-}d} I_k^{sev}$$

These models add more states to create a more elaborate and realistic simulation like:

1. **Infected states** :
   a. Asymptomatic
   b. Symptomatic
   c. Severe
2. **Dead state**

This framework allows a detailed study of spread, severity, and the impact of different kinds of infections done by the same virus.

Extensions of the SEIR model

Dashboard Applications

The dashboard from this research can be useful in a number of fields such as :

1. **Healthcare Resource Planning**:
   Predict hospitalizations and resource needs (e.g., ICU beds, ventilators) over time for better management

2. **Public Education and Communication**:
   Serve as an interactive tool to explain COVID-19 dynamics to the public or students.

3. **Research and Scenario Analysis**:
   Allow researchers to test various scenarios by adjusting parameters such as transmission rates or immunity waning.

Future Scope

This project can be further extended to analyse and simulate the progress of COVID-19 and other infectious diseases in different situations :

- Extend to include different at-risk groups
- Use more external factors like vaccination rates, mobility of public and implementation of government policies (lockdown, closing of schools).

These directions can give us more comprehensive and realistic results.

The dashboard can also be further improved by including more information about global safety measures and related data.

Resources

Data: covariants data
Codebase: Github Repo
ECDC
Wiki-Variants-of-SARS-CoV-2
covariants.org
cov-lineages.org
SARS-CoV-2 journey: from alpha variant to omicron and its sub-variants | Infection
Alpha, Beta, Gamma, Delta, Omicron: COVID-19 Variants Explained
From Alpha to Omicron: How Different Variants of Concern of the SARS-Coronavirus-2 Impacted the World - PMC
WHO COVID-19 dashboard
GISAID - hCoV-19 Variants Dashboard
Nextstrain / ncov / gisaid / global / 6m
COVID-19 - ETL
Extensions of the SEIR model

THANK YOU

Questions?

Our Project Scope

- Analysis on Effect and Damage(death) by different variants.

  ———-
- Studying the timeline of different variants(1)
- Understanding seasonality and forecasting for the different variants(6-7)
- Overview of mutations leading to variants(1 heatmap)
- SE I^3 RD Simulation - Dashboard

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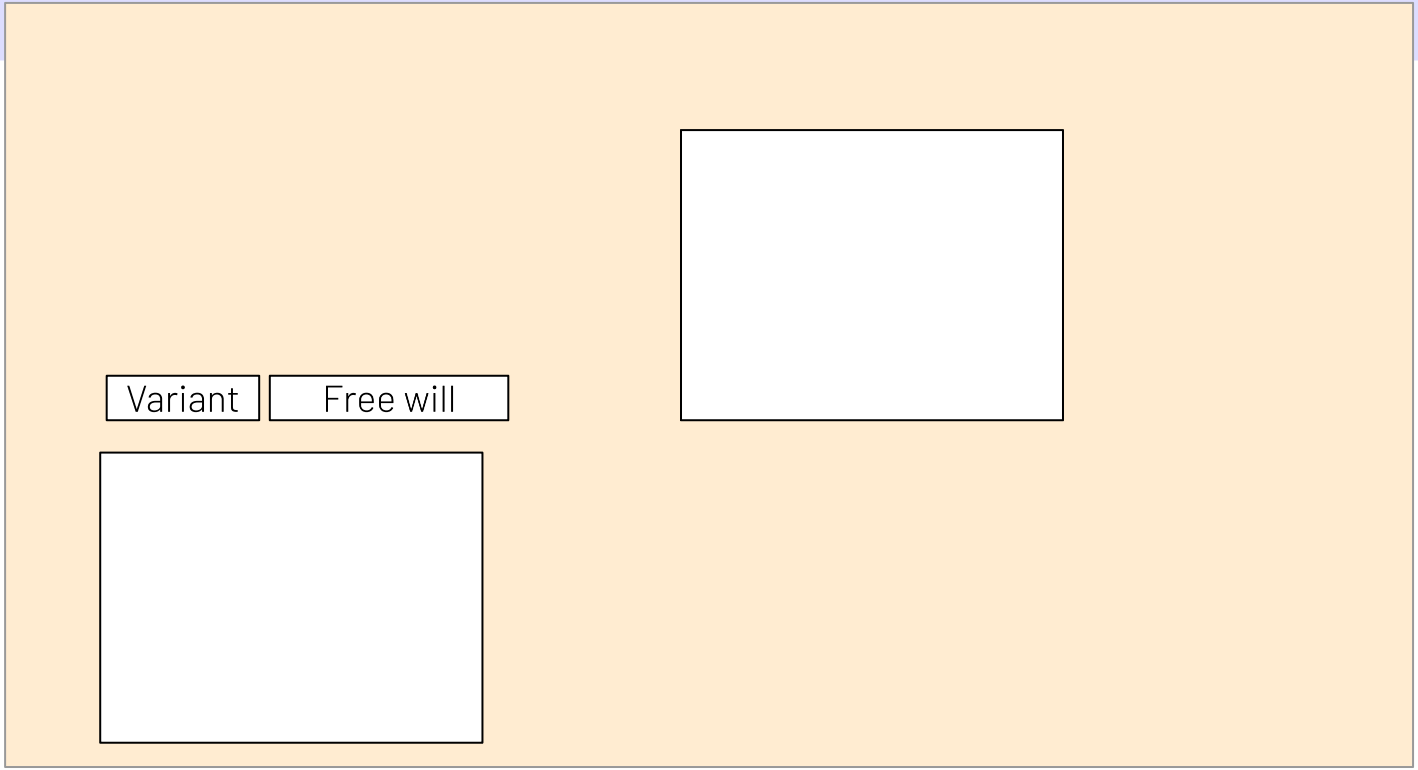
Dashboard



REVENUE

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Questions?

Questions?

Questions?

MARKET TRENDS

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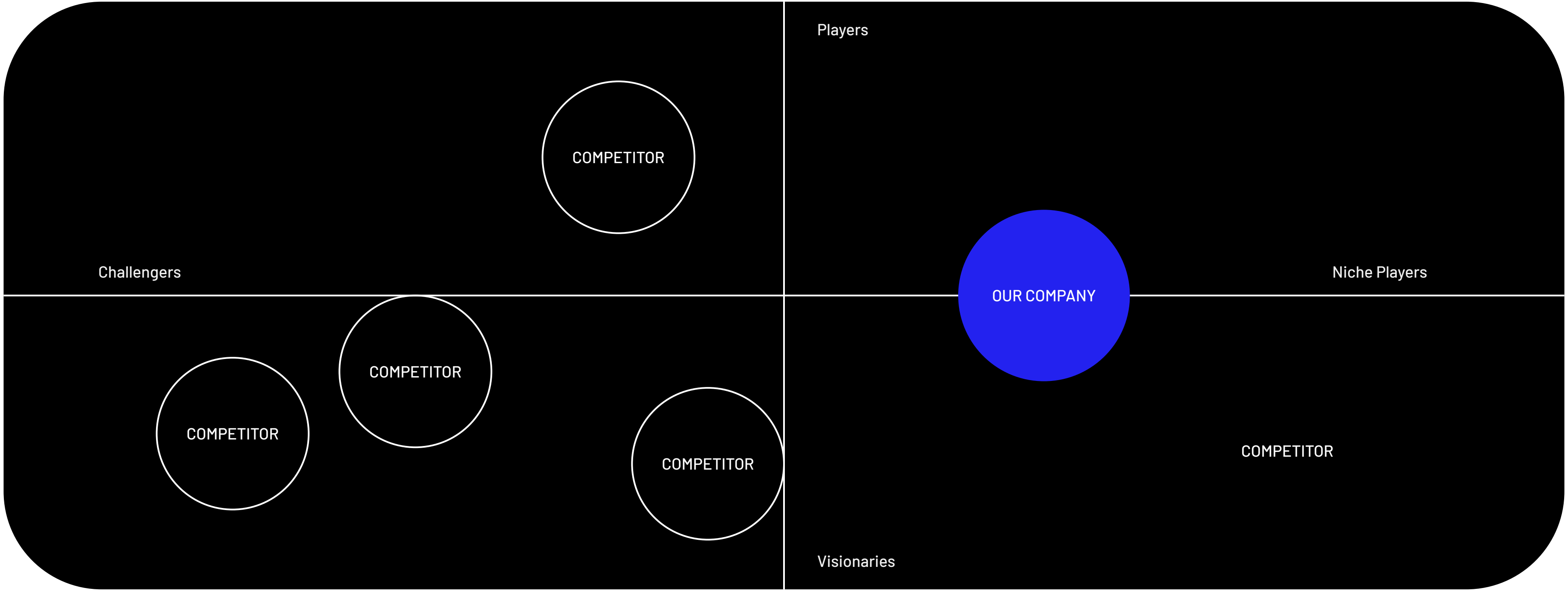

Questions?

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